Sample OSPF Configuration Between Two Routers

Network Topology

LAN 1 WAN Link LAN 2

192.168.1.0/24 10.0.0.0/30 192.168.2.0/24

PC —— R1 ———————————— R2 —— PC

Router 0

enable

configure terminal

router ospf 1

network 192.168.1.0 0.0.0.255 area 0

network 10.0.0.0 0.0.0.3 area 0

end

write memory

Router 1

enable

configure terminal

router ospf 1

network 192.168.2.0 0.0.0.255 area 0

network 10.0.0.0 0.0.0.3 area 0

end

write memory

Configuration Line	Explanation
`enable`	Enters privileged EXEC mode so administrative commands can be used.
`configure terminal`	Switches to global configuration mode to configure the router.
`router ospf 1`	Starts the OSPF routing process with Process ID **1**. The process ID is locally significant and identifies the OSPF instance on the router.
`network 192.168.1.0 0.0.0.255 area 0`	Enables OSPF on the **192.168.1.0/24** LAN network and advertises it in **Area 0**.
`network 10.0.0.0 0.0.0.3 area 0`	Enables OSPF on the WAN link between Router 1 and Router 2 so they can exchange routing information.
`end`	Exits configuration mode and returns to privileged EXEC mode.
`write memory`	Saves the running configuration to startup configuration so it remains after a reboot.

How OSPF Works in This Example

1. Router 1 and Router 2 enable OSPF using **Process ID 1**.
2. Both routers enable OSPF on the shared **10.0.0.0/30** WAN link.
3. The routers discover each other and form an **OSPF neighbor relationship**.
4. They exchange routing information.
5. Router 1 learns about the **192.168.2.0/24** network, and Router 2 learns about the **192.168.1.0/24** network.
6. Devices on both LANs can now communicate automatically without adding static routes.

Final Output

In this configuration, OSPF is enabled on both routers using **Process ID 1**. Each router advertises its local LAN network and the shared WAN link in **Area 0**, which is the backbone area of OSPF. After the routers become OSPF neighbors, they automatically exchange routing information. This allows Router 1 to learn the route to **192.168.2.0/24** and Router 2 to learn the route to **192.168.1.0/24**, enabling communication between the two networks without manually configuring static routes.